

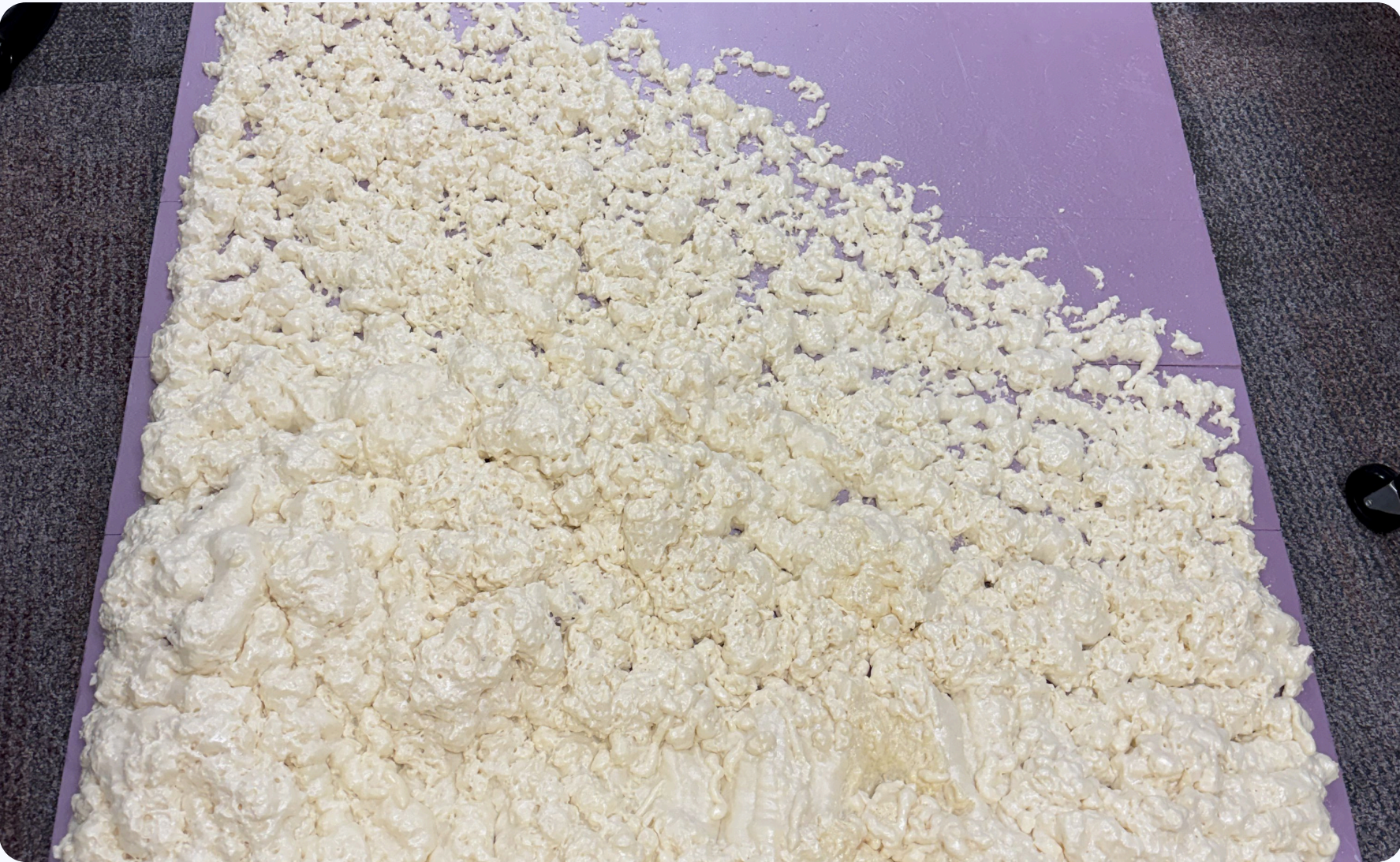
MISSION OVERVIEW

Design a multi-rover system for lunar and Martian exploration, paired with a 4×4 ft simulation landscape for NASA's HERA / Artemis II program.

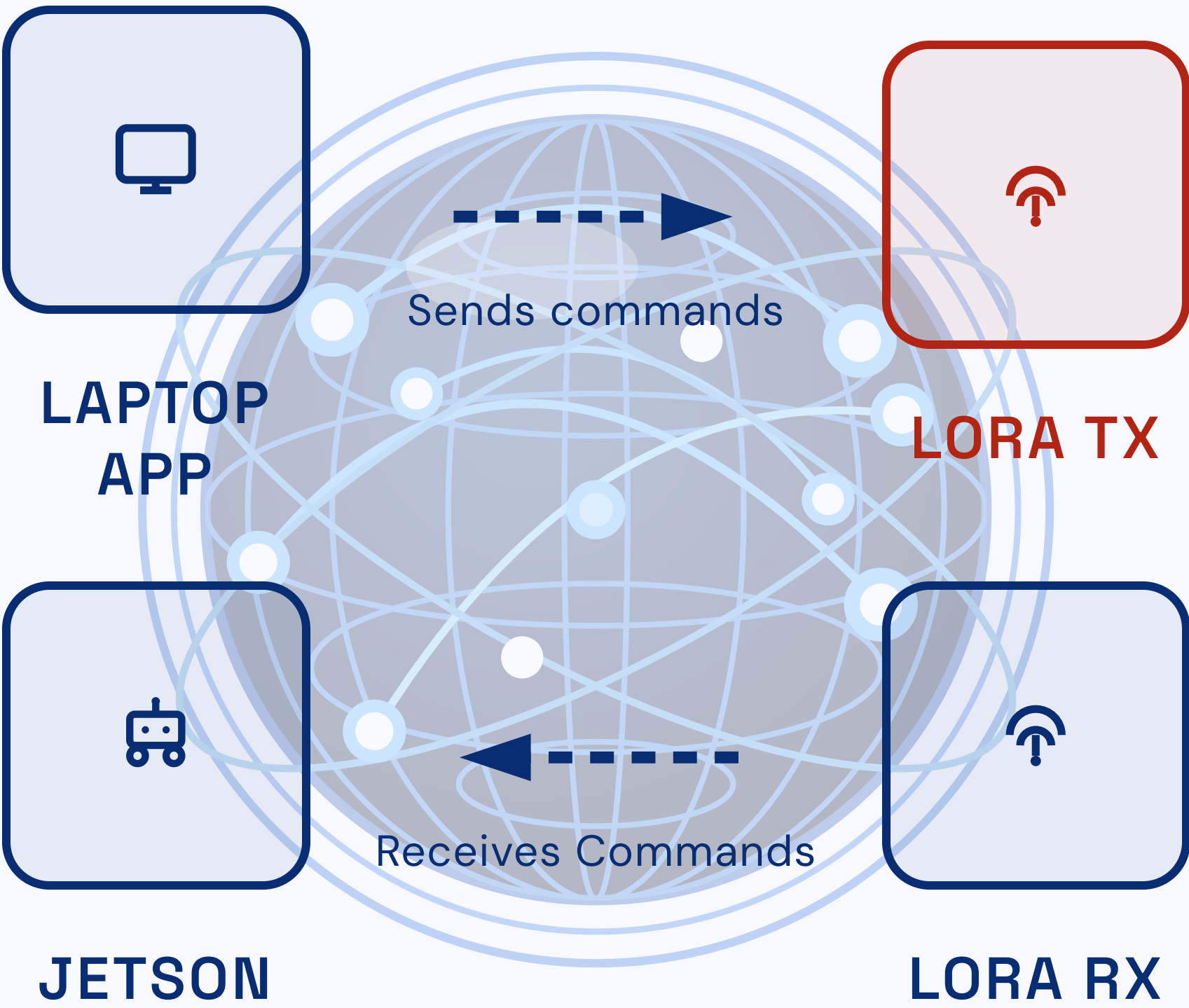
- **Rover Swarm** Three specialized rovers for arm ops, terrain scouting, and support
- **Terrain** 4×4 ft dual-terrain landscape for realistic rover testing
- **Crew Immersion** Simulates authentic extraterrestrial surface conditions

TERRAIN

Foamboard, foam gap filler, and diatomaceous earth simulate regolith. Modular 4×4 ft sections for transport & reconfiguration.



LORA SIGNAL PATH



ASTRONAUT APP



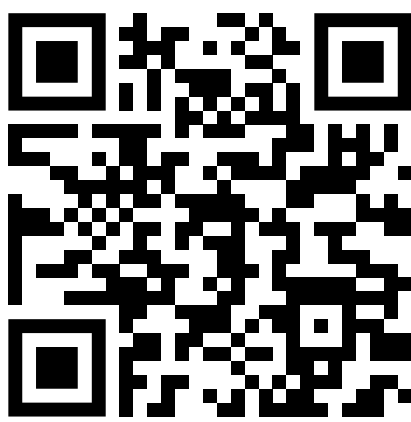
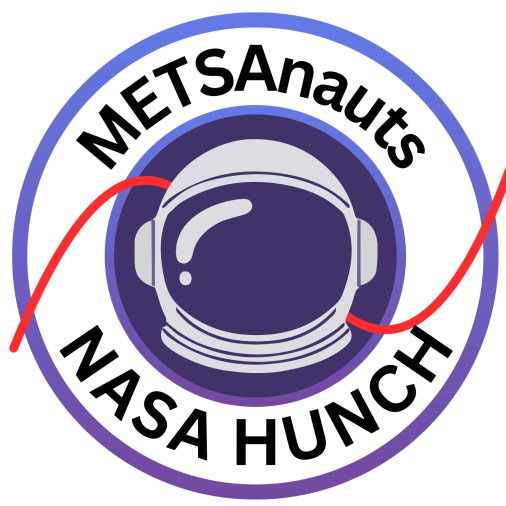
An easy-to-use, AI-driven communication app built for astronauts. No technical training required.

ROVER SWARM OVERVIEW

- **Claw Rover:** Equipped with a robotic arm for sample collection and manipulation.
- **Recon Rover:** Focused on terrain mapping, sensor coverage, and communication relay.
- **Support Rover:** Assists with logistics, redundancy, and swarm coordination.
- **All Rovers:** WiFi-primary with automatic LoRa fallback beyond signal range.

SPATIAL MAPPING

- ZED 2i stereo cameras mounted on the front of both support rovers
- Provides real-time depth sensing using 3D data
- Dual-camera setup creates overlapping fields of view across the swarm
- Keeps obstacle detection covered across the full operating area
- Makes cubesat detection and handling more reliable



ROVER SYSTEMS

CORE ARCHITECTURE

- **Rocker Bogie Suspension:** Keeps body upright over rough terrain
- **LORA communication:** Long-range radio communication, ~1km
- **6 Independent Motors:** One per wheel for precise torque control
- **Jetson Nano:** Central control with WiFi and LoRa interfaces
- **ZED 2i Camera:** Live stereo video feed and real-time spatial mapping

MOBILITY

- 6-wheel drive distributes weight on uneven terrain
- High ground clearance for rock obstacles
- Front climbs while rear anchors on inclines

ROBOTIC CLAW

- Collects soil and rock samples
- Camera-guided precision placement
- WiFi or LoRa control by range

KEY INNOVATIONS

SPATIAL MAPPING

ZED 2i stereo camera generates real-time 3D maps of surrounding terrain.

LORA FALLBACK

Reliable ~1 km backup link when WiFi range is exceeded.

ASTRONAUT UI

AI-powered web app. No coding or training required.

HALF-TREAD WHEELS

Wider contact prevents sinking on loose, sandy regolith.

LAN NETWORK

Laptop-hosted network. No internet required.

POWERED SUSP.

Body angle adjusts dynamically for terrain or sun position.

COMMUNICATION STACK

WiFi

Main connection for close-range control and claw camera feed (works up to ~20 ft)

LoRa

Backup for long-distance operation, reaching up to ~1 km; sends commands as hex that gets decoded automatically

LAN

Runs off a laptop hotspot so the rover connects directly — no internet required

APP

Web-based control interface with AI built in, simple enough for anyone on the crew to pick up